
From Inverse Optimization to Sequential Decisions: Conformal Prediction for Inverse Reinforcement Learning

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Abstract

1 Inverse reinforcement learning (IRL) is inverse optimization with a sequential
2 forward problem: demonstrations reveal an unknown reward, which is then op-
3 timized to prescribe a policy. Seen this way, IRL inherits a weakness that Con-
4 formal Inverse Optimization (CIO) identified for one-shot decisions – collapsing
5 a heterogeneous population of demonstrators to one reward point estimate can
6 produce an arbitrarily bad policy. We introduce **CP-IRL**, which transfers CIO’s
7 calibrate-then-robustify principle to Markov decision processes: from a population
8 of demonstrators, CP-IRL conformally calibrates a set of reward parameters con-
9 sistent with a new demonstrator, then optimizes a policy against that ambiguity.
10 Three pieces of new machinery make this tractable. First, the Bellman resolvent
11 reduces each calibration score to a reward-dimensional (rather than reward-plus-
12 value-function-dimensional) convex program. Second, we show the robust MDP is
13 convex whenever ambiguity is confined to the reward and the transition kernel is
14 fixed – for any compact ambiguity set – and that this boundary is tight: a two-state
15 counterexample loses convexity the moment ambiguity also touches the dynamics.
16 Third, for the calibrated cap specifically, the robust MDP collapses exactly into the
17 point-estimate MDP plus an ℓ_2 penalty on the feature component orthogonal to the
18 estimated reward, with penalty weight $\tan \alpha_\gamma$ set by the conformal quantile itself
19 – calibration is thus a principled way to tune a regularizer, and the reformulation
20 predicts when robustification can help at all. Building on this machinery, we prove
21 finite-sample coverage, a regret bound, and a separation result where point-estimate
22 IRL’s regret grows without bound while CP-IRL’s vanishes. Experiments on ran-
23 dom MDPs, gridworld, and Objectworld then confirm the predicted scope: robust
24 hedging cuts true-reward regret under systematic estimation bias, but costs regret
25 when the point estimate is merely noisy and already unbiased.

1 Introduction

26 Inverse reinforcement learning (IRL) is usually introduced as its own problem: infer a reward from
27 behavior, then optimize that reward to get a policy. But end to end, IRL is a sequential instance of
28 *inverse optimization* – the observed decision is a policy or trajectory, the latent objective is a reward
29 function, and the forward problem is an MDP. Under that view, uncertainty about the recovered reward
30 should propagate into the downstream decision instead of being discarded once a point estimate is fit.

31 Conformal Inverse Optimization (CIO) builds exactly this pipeline for one-shot decisions Lin et al.
32 [2024]: it replaces a single recovered objective with a conformally calibrated uncertainty set and
33 optimizes robustly over that set, with a finite-sample, distribution-free guarantee under exchangeabil-
34 ity. CIO’s own separation example shows the stakes – point-estimate inverse optimization can carry
35 unbounded downstream regret even with unlimited data. This raises the question we answer here:
36

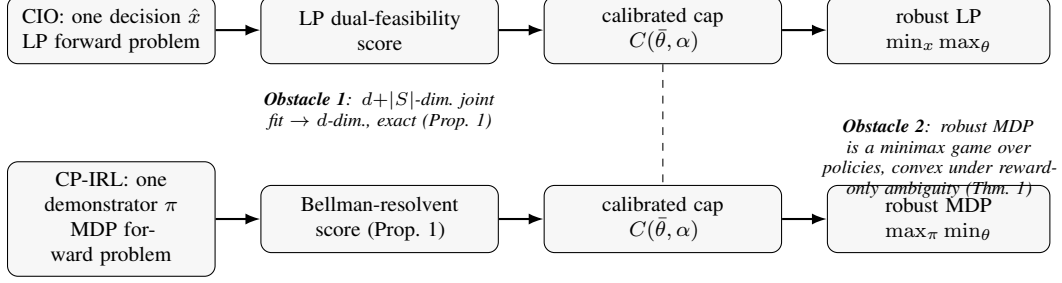


Figure 1: The transfer at a glance. CP-IRL keeps CIO’s calibrate-then-robustify template (same calibrated cap $C(\theta, \alpha)$, dashed link) but replaces every sequential-specific piece: the nonconformity score (§4.1) and the robust decision problem (§4.2).

can CIO’s calibrate-then-robustify principle transfer to inverse reinforcement learning without losing coverage or tractability?

Our answer, CP-IRL, keeps CIO’s pipeline and rebuilds its machinery for sequential decisions. A population of demonstrators supplies the exchangeable calibration units; each observed policy defines a feasible reward set through Bellman optimality; split conformal prediction calibrates a spherical cap around an IRL point estimator; a robust MDP turns the calibrated ambiguity into a deployed policy. Two obstacles block a direct transplant of this template from LPs to MDPs.

Obstacle 1: the calibration score is no longer a duality fact. CIO’s per-point score is computable in closed form because the forward problem is a linear program, so reward feasibility reduces to LP dual feasibility. An MDP has no such shortcut: “this policy is optimal for this reward” is a Bellman-optimality condition, and checking it naively means jointly optimizing over the reward and the policy’s value function, a variable count that grows with the state space.

Obstacle 2: the robust decision is a game over policies. CIO’s robust forward problem is a single robust linear program, but the MDP analogue is a minimax game, $\max_{\pi} \min_{\theta}$, over an ambiguity set – and robust MDPs with ambiguity that couples across states are NP-hard in general Wiesemann et al. [2013], with tractable results typically relying on *rectangularity* assumptions that restrict how the adversary can vary its choice by state.

We resolve both obstacles, and then show that resolving the second one buys more than tractability: for the calibrated cap, the robust MDP is *exactly* a regularized point-estimate MDP, whose penalty weight the conformal quantile sets on its own (Proposition 2). That reformulation is what lets us predict, rather than merely report, when robustification pays off. Figure 1 lays out the transfer: the same calibrate-then-robustify template on both rows, with the two obstacles resolved where the sequential setting departs from CIO’s one-shot LP. The contributions are:

- **Transfer: conformal inverse optimization for IRL.** We formulate one demonstrator as one exchangeable inverse-optimization instance, calibrate a reward uncertainty set through Bellman-optimal reward feasibility, and establish finite-sample coverage together with a true-reward regret bound (§3–§5).
- **Optimization: sequential structure without state-sized calibration** (§4.1). For a fixed demonstrator policy, the Bellman resolvent $(I - \beta P_{\pi})^{-1}$ makes the induced value function *linear* in the reward parameter – the same identity underlying Ng & Russell’s Ng and Russell [2000] feasible-reward characterization, which we repurpose as a calibration-score reduction. Substituting it shrinks the per-demonstrator calibration score from a joint optimization over (θ, V) (size $d + |S|$) to one over θ alone (size d), exactly, by the policy improvement theorem. We also give a Sherman–Morrison–Woodbury update that recomputes the resolvent in $O(k^2|S| + k^3)$ rather than $O(|S|^3)$ when two demonstrators’ policies differ at only k states.
- **Tractability: a boundary between reward and dynamics ambiguity** (§4.2). The robust MDP is a convex program in occupancy-measure space for any compact reward ambiguity set, with the transition kernel fixed – a Bellman-recursion-free argument that extends the closest sufficiency-side result (Gadot et al. Gadot et al. [2024], who solve the convex L_p -ball case via Hölder duality and policy gradient) to arbitrary compact sets. We show this boundary

is tight: a two-state, two-action MDP with a one-dimensional ambiguity set coupling reward and transition probability breaks concavity, with a hand-verifiable rational-valued Jensen violation.

- **Interpretation: conformal calibration is regularizer tuning** (§4.3, Proposition 2). For the calibrated cap, the robust MDP reduces in closed form to $\max_{\mu} \{\bar{\theta}^\top \Phi^\top \mu - \tan \alpha_\gamma \|\Phi^\top \mu - (\bar{\theta}^\top \Phi^\top \mu) \bar{\theta}\|_2\}$: the point-estimate MDP, minus an ℓ_2 penalty on the expected-feature component orthogonal to the estimated reward direction. The conformal quantile sets the penalty weight $\tan \alpha_\gamma$ with no hyperparameter to choose, and the penalty is a *deterministic* function of $\bar{\theta}$ alone – which is precisely why it can correct a systematic error but not an unbiased one.
- **Scope: robustness helps under bias, not noise alone.** Our sequential analogue of CIO’s separation example proves classical point-estimate IRL’s regret can diverge while the calibrated policy’s regret vanishes. Across random MDPs, a region-based gridworld, and Objectworld, calibrated robust hedging reduces true-reward regret when the point estimate is *systematically* biased (as our degenerate construction manufactures), and increases it when the estimate is merely noisy but unbiased – exactly the boundary Proposition 2 predicts, and one we report as a core finding rather than bury as a limitation.

2 Related Work

Inverse optimization and its calibration. Inverse optimization recovers parameters of an optimization problem from observed decisions (Ahuja & Orlin Ahuja and Orlin [2001]; Chan et al. Chan et al. [2018]). Classical formulations return a point estimate; more recent work constructs data-driven uncertainty sets around it (Chenreddy, Bandi & Delage Chenreddy et al. [2022]; Sun, Liu & Li Sun et al. [2023]), but – as Lin, Delage & Chan Lin et al. [2024] show – none of these give the specific guarantee we need: that the *next* observed decision’s true parameter is covered with a stated confidence, distribution-free. Conformal Inverse Optimization is the first to close this gap using split conformal prediction Vovk et al. [2005], and is the paper this work directly extends from one-shot decisions to MDPs.

Feasible reward sets in IRL. A separate line of work characterizes the *set* of rewards consistent with observed IRL data (Poiani et al. Poiani et al. [2024]; Lazzati & Metelli Lazzati and Metelli [2025]; Skalse et al. Skalse et al. [2022]) and proves PAC-style bounds on recovering it, using it directly as a robust-MDP ambiguity set. This line shares our target object – a feasible reward set characterized by Bellman-optimality constraints – but its guarantees are PAC bounds requiring a generative model of demonstrator sub-optimality; ours require only exchangeability of demonstrators, no assumption on the *process* by which any individual demonstrator deviates from optimality. The transfer question this line raises – whether a feasible reward set identified on one MDP transfers to another (Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024]) – is a genuinely open problem for our setting that we do not attempt to close here (see §7).

Robust MDPs and the rectangularity boundary. Robust MDPs with ambiguity coupling across states are NP-hard in general (Wiesemann, Kuhn & Rustem Wiesemann et al. [2013]); tractability is classically recovered via *s*- or *sa*-*rectangularity* (Iyengar Iyengar [2005]; Nilim & El Ghaoui Nilim and El Ghaoui [2005]), which restricts the adversary to choose independently per state. Recent work extends tractable, *non*-rectangular ambiguity specifically for reward-only uncertainty with a fixed transition kernel: Gadot et al. Gadot et al. [2024] solve the convex L_p -ball case exactly via Hölder duality and a policy-gradient method with proven convergence; Xu et al.’s Group-Robust MDPs Xu et al. [2026] handle *feature-wise* (d-rectangular) ambiguity that couples both reward *and* transition, at the cost of restricting to that rectangular structure. Our sufficiency result (§4.2) covers the same fixed-kernel, reward-only regime as Gadot et al. but for an *arbitrary* compact ambiguity set (not only L_p -balls), via a direct occupancy-measure convex program rather than a policy-gradient method; our necessity result – an explicit demonstration that convexity is lost the moment ambiguity couples into the kernel, even minimally – has no counterpart in either paper.

Conformal prediction in RL-adjacent settings. Conformal prediction has been used for offline-RL pessimism from Q-value or state-action-coverage uncertainty (UARM, Pan et al. Pan et al. [2026]), for calibrated intervals in infinite-horizon policy evaluation, and for modeling *other agents’ actions* as conformal prediction sets in multi-agent RL (CAMMARL, Gupta & Kahou Gupta and Kahou [2023]) – the last of these is explicitly motivated by an analogy to IRL but never calibrates a reward

set. None combine conformal calibration with a reward vector recovered from a population of IRL demonstrators and a robust-MDP regret guarantee, which is the gap this paper closes.

Preference learning and RLHF. Preference-based and RLHF-style learning fits a reward (or policy) from labeled comparisons or demonstrations, typically from individual expert trajectories (Christiano et al. Christiano et al. [2017]; Rafailov et al. Rafailov et al. [2023]) or crowdsourced preference populations. Our per-demonstrator exchangeable-unit design (§3) is structurally suited to recast “one annotator, one calibration point” for heterogeneous-preference RLHF populations. The resulting coverage guarantee is in the intersection-nonemptiness sense of Theorem 2, not parameter containment: the calibrated reward-ambiguity set is compatible with a γ -fraction of the annotator population’s individual reward functions, which are never directly observed. We are not aware of this guarantee elsewhere in the RLHF calibration literature, though we test it only on a toy response-selection MDP, not at RLHF’s actual scale (§7).

3 Setting

We now make the CIO–IRL correspondence precise. We adopt the population-of-demonstrators setting used throughout this work: a shared tabular MDP $M = (S, A, P, \beta, \phi)$ with known transition kernel P , discount β , and feature map $\phi : S \times A \rightarrow \mathbb{R}^d$ inducing reward $r_\theta(s, a) = \phi(s, a)^\top \theta$. A population of demonstrators draws perceived rewards $\hat{\theta}_1, \dots, \hat{\theta}_N \sim P_\theta$ i.i.d. and each acts optimally in M under their own $\hat{\theta}_k$, producing an observed policy $\pi_k = \pi^*(\hat{\theta}_k)$ – we assume the full deterministic policy is observed exactly, not estimated from a finite trajectory sample; relaxing this to trajectory-level observation would make $\hat{\Theta}(\pi_k)$ itself an estimated (rather than exact) feasible set and is outside the scope of the guarantees below.

Assumption 1 (Exchangeable calibration unit). *The pairs $(\hat{\theta}_k, \pi_k)$, $k = 1, \dots, N, N+1$, are i.i.d. (hence exchangeable), and M (in particular P) is shared and fixed across all demonstrators and the deployment instance.*

Under Assumption 1, each demonstrator plays the role of one (decision, context) pair in CIO’s setting, and it is the setting under which CIO’s coverage argument transplants essentially verbatim (§5, Theorem 2). We commit to this “unit A” design over two weaker alternatives – per-task exchangeability with a single demonstrator (still exchangeable, treated as an easy corollary) and per-trajectory exchangeability within one demonstrator (broken by within-trajectory Markov dependence, and used only as an ablation) – and are explicit throughout about what breaks under each relaxation.

The learning task, mirroring CIO’s pipeline, has three steps: (i) split the population into training and calibration sets; (ii) fit a point estimate $\bar{\theta}$ on the training set with any IRL method (we use and compare a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL by MCMC – §6); (iii) calibrate a spherical-cap uncertainty set

$$C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \bar{\theta} \geq \cos \alpha\}$$

on the calibration set and solve the robust MDP

$$\max_{\pi} \min_{\theta \in C(\bar{\theta}, \alpha)} \theta^\top \mu(\pi)$$

in occupancy-measure space, where $\mu(\pi)$ is π ’s occupancy measure.

4 Method

4.1 Tractable calibration via the Bellman resolvent

CIO’s calibration score for a validation point k is

$$c_k = \max_{\bar{\theta}} \{\theta^\top \bar{\theta} : \theta \in \Theta^{\text{OPT}}(\hat{x}_k, u_k), \|\theta\|_2 \leq 1\},$$

where Θ^{OPT} is characterized by LP dual feasibility – tractable because the forward problem is a linear program. In an MDP, the analogous feasible set for a policy π is

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \forall s, \forall a\},$$

the exact set of reward vectors for which π is Bellman-optimal, by the policy improvement theorem. Computing this naively requires the joint optimization $\max_{\theta, V} \theta^\top \bar{\theta}$ subject to the Bellman-optimality inequalities written in terms of both θ and the a priori unknown value function V – a variable count of $d + |S|$.

The key fact we use, which is the sequential-decision restatement of the feasible-reward characterization of Ng & Russell Ng and Russell [2000]: for a **fixed** policy π , the policy’s own induced value function V_θ^π is *linear* in θ , via the Bellman resolvent.

Proposition 1 (Resolvent-based score reduction). *Fix a policy π , let $P_\pi \in \mathbb{R}^{|S| \times |S|}$ be the induced state-transition matrix ($P_\pi(s, s') = P(s, \pi(s), s')$), and let $\Phi_\pi \in \mathbb{R}^{|S| \times d}$ be the feature matrix restricted to the actions π takes ($\Phi_\pi(s, \cdot) = \phi(s, \pi(s))^\top$). Let*

$$W_\pi := (I - \beta P_\pi)^{-1} \Phi_\pi$$

(well-defined since $\beta \in [0, 1)$).

Then $V_\theta^\pi = W_\pi \theta$ and $Q_\theta^\pi(s, a) = \Psi_\pi(s, a)^\top \theta$ with

$$\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot)^\top W_\pi,$$

both linear in θ .

Consequently,

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \ \forall s, a\},$$

exactly (no relaxation), and the calibration score

$$c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$$

is an optimization over θ alone – size d , not $d + |S|$ – with W_{π_k} (and hence Ψ_{π_k}) precomputed once per distinct policy via a single resolvent solve.

The proof substitutes the linear Bellman equation’s solution into Q_θ^π and rewrites the policy-improvement optimality condition using Ψ_π ; full proof in Appendix A.1. We checked Proposition 1 against a slow joint- (θ, V) reference implementation and against independent value iteration, matching to solver precision on randomized instances; the speedup over the joint formulation grows with $|S|$, 2–5.5 $\times$ at a few hundred states. The Sherman–Morrison–Woodbury update – recomputing the resolvent in $O(k^2|S| + k^3)$ rather than $O(|S|^3)$ when a new demonstrator’s policy differs from an already-resolved one at only k states – has no counterpart in Ng & Russell Ng and Russell [2000]; it is the piece specific to calibrating against a whole population rather than a single demonstrator.

4.2 A tight tractability boundary for the robust MDP

The robust decision problem,

$$\max_{\pi} \min_{\theta \in C} \theta^\top \mu(\pi),$$

is a minimax game over policies. We work in occupancy-measure space: $M(P)$ denotes the (fixed, since P is known) polytope of achievable occupancy measures μ , normalized so $\sum_{s,a} \mu(s, a) = 1$, and $\Phi \in \mathbb{R}^{|S| \times A| \times d}$ is the matrix whose rows are $\phi(s, a)^\top$ (matching the convention already used for Φ_π in §4.1), so that

$$\theta^\top \Phi^\top \mu = \sum_{s,a} \mu(s, a) r_\theta(s, a)$$

is the expected per-step reward of occupancy measure μ under reward θ , and $\theta^\top \Phi^\top \mu / (1 - \beta)$ is the corresponding expected discounted return. The robust objective is

$$\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$$

(the $1/(1 - \beta)$ factor is a positive constant and does not affect the $\arg \max$).

206 **Theorem 1** (Reward-only ambiguity is convex; this is tight). *(a) Sufficiency.* For any nonempty
 207 compact ambiguity set $C \subseteq \mathbb{R}^d$ (e.g., our calibrated spherical cap $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \bar{\theta} \geq$
 208 $\cos \alpha\}$, a cap of the unit ball rather than the unit sphere), if the transition kernel P is fixed and known,
 209 then

$$\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$$

210 is a concave maximization over the fixed compact polytope $M(P)$, hence a convex program.

211 *(b) Necessity.* This can fail as soon as ambiguity couples reward and transition probability, even min-
 212 imally: there exists a 2-state, 2-action MDP and a convex, one-dimensional, non-product ambiguity
 213 set over (θ, P) jointly for which the analogous worst-case value function is not concave.

214 Part (a) holds because $g_C(x) := \min_{\theta \in C} \theta^\top x$ is concave regardless of C 's shape and $M(P)$ does
 215 not depend on θ ; part (b) exhibits an explicit rational-valued Jensen violation. Full proof, including
 216 the closed-form counterexample, in Appendix A.2. Part (a) generalizes the sufficiency result of Gadot
 217 et al. Gadot et al. [2024] (who solve the convex L_p -ball case exactly via Hölder duality and policy
 218 gradient) to an arbitrary compact C , via a direct occupancy-measure argument. Part (b) shows the
 219 coupling required to break concavity is minimal – a single scalar t linking reward and transition
 220 ambiguity along a line segment, far short of a fully unconstrained non-rectangular adversary – and, to
 221 our knowledge, is not established by either Gadot et al. or Xu et al. Xu et al. [2026]. Note that failure
 222 of concavity is a failure of this *particular* tractability argument, not by itself a proof of NP-hardness;
 223 general non-rectangular robust MDPs are already known to be NP-hard Wiesemann et al. [2013].

224 4.3 What the robust MDP is actually doing: a calibrated regularizer

225 Theorem 1 says the robust program is solvable, but not what it computes. For our calibrated cap the
 226 inner minimization has a closed form, and it turns the whole minimax problem into something far
 227 more familiar.

228 **Proposition 2** (The calibrated robust MDP is a regularized MDP). *Let $\|\bar{\theta}\|_2 = 1$ and $\alpha \in [0, \pi/2)$.
 229 For $x \in \mathbb{R}^d$ write the decomposition along and across the estimated reward direction, $x = a(x)\bar{\theta} + x_\perp$
 230 with $a(x) := \bar{\theta}^\top x$ and $x_\perp \perp \bar{\theta}$, and set $b(x) := \|x_\perp\|_2$. Then the inner worst case over the cap is*

$$\min_{\theta \in C(\bar{\theta}, \alpha)} \theta^\top x = \begin{cases} -\|x\|_2, & \text{if } a(x) \leq -\|x\|_2 \cos \alpha, \\ a(x) \cos \alpha - b(x) \sin \alpha, & \text{otherwise.} \end{cases}$$

231 Consequently, whenever every $\mu \in M(P)$ earns nonnegative estimated reward ($a(\Phi^\top \mu) \geq 0$, e.g.
 232 when $\bar{\theta}^\top \phi(s, a) \geq 0$ for all (s, a)), the second branch applies throughout and the robust policy solves

$$\mu_{\text{CIRL}} \in \arg \max_{\mu \in M(P)} \left\{ \underbrace{\bar{\theta}^\top \Phi^\top \mu}_{\text{point-estimate objective}} - \tan \alpha \cdot \underbrace{b(\Phi^\top \mu)}_{\text{penalty}} \right\}.$$

233 So CP-IRL is the point-estimate MDP with an ℓ_2 penalty on how far the achieved expected-feature
 234 vector points away from the estimated reward direction, at weight $\tan \alpha$. (This also re-proves
 235 Theorem 1(a) constructively for this C : a linear term minus a seminorm of a linear image of μ is
 236 visibly concave.) The $\tan \alpha$ form needs $\alpha < \pi/2$; a calibrated $\alpha_\gamma \geq \pi/2$ would mean a cap spanning
 237 more than a hemisphere, at which point the set is essentially vacuous anyway. Every α_γ we calibrate
 238 in §6 is well inside this range. Two consequences matter for the rest of the paper.

239 **Calibration is regularizer tuning.** The penalty weight is not a hyperparameter but $\tan \alpha_\gamma$, fixed by
 240 the conformal quantile of Theorem 2. The limits behave correctly: $\alpha_\gamma \rightarrow 0$ (a population that does
 241 not disagree) recovers the plain point estimate, and $\alpha_\gamma \rightarrow \pi/2$ (a vacuous cap) sends the weight to
 242 $+\infty$, refusing any occupancy measure with an off-direction feature component. Choosing a target
 243 coverage γ is thus, precisely, choosing a regularization strength that arrives with a finite-sample
 244 guarantee attached.

245 **The penalty cannot see the estimation error it would need to cancel.** It depends on the data
 246 only through $\bar{\theta}$ and α_γ , so given those it is a deterministic tilt of the objective carrying no further
 247 information about θ_{new}^* . It can therefore improve true-reward value only when it happens to undo
 248 an error already fixed in $\bar{\theta}$; against unbiased noise it is uncorrelated with the error it must cancel

and, in expectation, can only cost value. This predicts the scope condition §6 measures. We offer it as the mechanism our experiments corroborate rather than as a theorem: proving it would require quantifying the tilt against a specified estimator-error distribution, which we do not attempt.

The same form also prices the hedge before deployment. Writing $\mu_{\text{plain}} \in \arg \max_{\mu \in M(P)} \bar{\theta}^\top \Phi^\top \mu$, and using that μ_{CIRL} wins on the penalized objective while μ_{plain} wins on the unpenalized one,

$$0 \leq \bar{\theta}^\top \Phi^\top \mu_{\text{plain}} - \bar{\theta}^\top \Phi^\top \mu_{\text{CIRL}} \leq \tan \alpha_\gamma \cdot b(\Phi^\top \mu_{\text{plain}}).$$

Every quantity on the right is computable at deployment time without ground truth. When the plain optimum’s expected feature vector nearly aligns with $\bar{\theta}$ the premium is near zero and robustification is close to a no-op; when it is large, CP-IRL is betting heavily that $\bar{\theta}$ is systematically wrong. Read honestly, this bounds the loss in *estimated* reward: it measures how far robustification will move the decision, not whether the move is beneficial.

5 Theory

Three more results follow, each the MDP/IRL analogue of a corresponding CIO theorem: coverage, a regret bound, and separation. We spot-check each computationally against the implemented solvers rather than trust the derivation alone.

Theorem 2 (Coverage). *Under Assumption 1, let $\tau = \lceil (N_{\text{val}} + 1)\gamma \rceil$, let $c_{(\tau)}$ be the τ -th largest calibration score among $c_1, \dots, c_{N_{\text{val}}}$ (Proposition 1), and set $\alpha_\gamma := \arccos(c_{(\tau)})$. Then*

$$\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma,$$

where $\hat{\Theta}_{\text{new}}$ is the new demonstrator’s feasible reward set.

This is an intersection-nonemptiness guarantee, not a containment guarantee: it does not by itself assert $\theta_{\text{new}}^* \in C(\bar{\theta}, \alpha_\gamma)$, only that the calibrated cap is compatible with some reward for which π_{new} is optimal.

The proof transplants CIO’s exchangeability argument line for line onto the scores of Proposition 1 in place of CIO’s LP dual scores (Appendix A.4). A marginal-coverage simulation against the actual calibration code (150 trials per (γ, N_{val}) cell) confirms coverage $\geq \gamma$ at every tested combination; finite demonstrator populations produce tied (atomic) calibration scores, the likely source of the mild over-coverage we observe beyond the rank argument’s $1/(N_{\text{val}} + 1)$ slack (§6).

Theorem 3 (Regret bound). *Let $\|\phi\|_{2,\infty} := \max_{s,a} \|\phi(s, a)\|_2$ and define*

$$\nu(\mu) := \frac{\|\Phi^\top \mu\|_2}{1 - \beta} \leq \bar{\nu} := \frac{\|\phi\|_{2,\infty}}{1 - \beta}$$

for any occupancy measure $\mu \in M(P)$ – a Lipschitz constant for $\theta \mapsto \frac{\theta^\top \Phi^\top \mu}{1 - \beta}$ in θ , via Cauchy–Schwarz. Let μ_{CIRL} , μ^* denote the occupancy measures of the robust policy and of the true optimum $\pi^*(\theta_{\text{new}}^*)$, respectively. Suppose:

(i) Almost-sure covering: the calibrated cap $C(\bar{\theta}, \alpha_1)$ satisfies $\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_1) \neq \emptyset) = 1$ for a fresh demonstrator – strictly stronger than Theorem 2’s γ -coverage, and trivially (but uninformatively) achievable by taking α_1 at its maximal value, i.e. C the whole unit ball;

(ii) Bounded feasible-set diameter: any two unit-norm rewards optimal for the same policy are within η of each other;

(iii) Population accuracy: the population mean reward is within σ of θ_{new}^* in ℓ_2 .

Then the **average optimality gap**

$$\text{AOG} := V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}})$$

(the true-reward regret of the robust policy relative to the in-hindsight optimum) satisfies

$$\text{AOG} \leq (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu^*) + (\eta + \sigma) \nu(\mu_{\text{CIRL}}),$$

and substituting the uniform bound $\bar{\nu}$ for both $\nu(\mu^*)$ and $\nu(\mu_{\text{CIRL}})$ gives the fully distribution-free corollary

$$\text{AOG} \leq (2 \sin \alpha_1 + 2\eta + 2\sigma) \bar{\nu}.$$

288 The proof telescopes three terms – Lipschitz continuity of $\theta \mapsto \theta^\top \Phi^\top \mu$, a chord-length bound on
 289 $C(\bar{\theta}, \alpha_1)$, and the robust policy’s minimax optimality – re-derived for the MDP’s $\max_\pi \min_\theta$ sign
 290 convention rather than transplanted from CIO’s $\min_x \max_\theta$ one (Appendix A.5). We checked $\bar{\nu}$ as a
 291 valid uniform Lipschitz constant against 200 randomized MDPs (zero violations) and confirmed the
 292 bound’s numeric validity directly on random instances.

293 This bound is loose, and we say so plainly. Cross-checked against the exact closed-form regret for our
 294 degenerate construction below (where the almost-sure-covering condition already holds at $\alpha_1 = 0$),
 295 the general bound’s looseness *grows* with the construction’s scaling parameter – from roughly $8\text{--}15\times$
 296 at the low end of the tested range to roughly $390\text{--}430\times$ at the high end, since the exact regret decays
 297 to zero while $\eta, \sigma, \bar{\nu}$ converge to nonzero constants. CIO’s own Example 1 bound stays within a fixed
 298 $2.3\text{--}4\times$ over the same kind of comparison, because its construction keeps a richer feasible region that
 299 does not saturate this way; ours grows.

300 **Theorem 4** (Separation). *There exists a single-state, two-action MDP (the sequential-decision*
 301 *analogue of CIO’s degenerate Example 1, with reward features the negation of CIO’s original cost*
 302 *features) and an IRL point estimator such that:*

- 303 • *the point-estimate solution converges to a degenerate direction*

$$\theta_u = \frac{(1, u)}{\sqrt{1 + u^2}} \neq \theta^*$$

304 *(a unit vector, hence on the boundary of the unit ball that $C(\bar{\theta}, \alpha)$ is a cap of) as the scaling*
 305 *parameter $u \rightarrow \infty$, giving classical point-estimate IRL true-reward regret that grows without*
 306 *bound in u ;*

- 307 • *while the calibrated robust policy’s regret is bounded and shrinks to zero over the same range.*

308 *The robust optimum is a genuine mixture, with exact closed form*

$$p^*(u) = \frac{u^2}{1 + u^2},$$

309 *independent of the calibration angle.*

310 Theorem 4 is proved via Sion’s minimax theorem applied to the mixture’s closed form (Appendix A.6),
 311 and independently re-verified against the actual implemented solver to 10^{-6} precision. Figure 2
 312 shows the live 10-seed computational confirmation. This separation exhibits the LP-based point
 313 estimator’s failure mode specifically (a degenerate boundary direction under tie-breaking); §6 Q3
 314 reports that MaxEnt and Bayesian IRL do *not* exhibit it on this same construction, and explains why.

315 6 Experiments

316 Three questions organize the experiments: (Q1) does calibration hit its target coverage without
 317 becoming vacuous; (Q2) when does robustification actually improve downstream regret; (Q3) do
 318 the conclusions hold beyond one IRL estimator or environment? We evaluate three point estimators
 319 (a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL via Metropolis–Hastings),
 320 each combined with three decision rules (the plain point estimate, our calibrated robust policy, and a
 321 “concentration baseline” whose radius comes from a DKW concentration bound instead of an exact
 322 conformal quantile), across three environments: random tabular MDPs, a region-based gridworld
 323 with Voronoi reward features, and Objectworld, the standard smooth-feature IRL benchmark.

324 The headline Q1–Q3 numbers use $|S| \leq 36, d \leq 5$, sized to run on a laptop (20 CPU cores, no GPU,
 325 no commercial solver). To check this scale is not load-bearing, we reran the full gridworld pipeline at
 326 $|S| = 225, d = 7\text{--}6.25\times$ more states, 2 more reward dimensions – and confirmed the pipeline stays
 327 under 35 seconds per demonstrator population up to $|S| = 400, d = 8$. Q2’s finding is unchanged at
 328 $|S| = 225$: conformal-robust regret still exceeds the plain estimate’s for all three estimators (LP-IRL
 329 $1.03 \rightarrow 3.19$, MaxEnt $1.01 \rightarrow 2.88$, Bayesian $0.06 \rightarrow 0.36$). We make no claim beyond tabular
 330 MDPs; function approximation and language-model settings are untested.

331 **Q1: coverage tracks the target, and the set is not vacuous.** Realized coverage tracks γ in every
 332 environment and estimator tested ($\sim 0.88\text{--}0.89$ at $\gamma = 0.8$ on the gridworld), and the calibrated

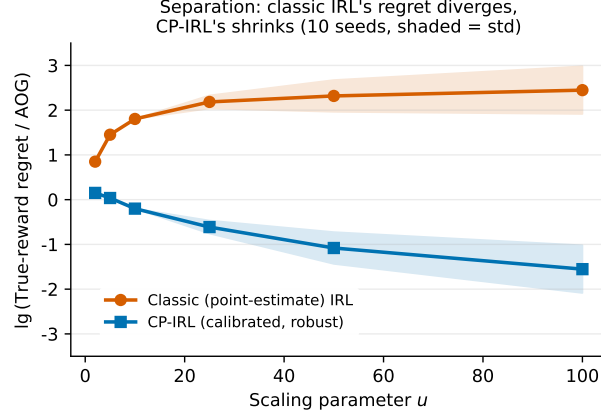


Figure 2: Separation result (Theorem 4): true-reward regret of the deployed policy, measured against the ground-truth reward θ_{new}^* , vs. the scaling parameter $u \in [2, 100]$ (log scale), 10 seeds, shaded regions show ± 1 std. “Classic IRL” is the sub-optimality-loss linear-program point estimator (§6), which converges to the degenerate direction θ_u ; its regret (orange) grows monotonically and without bound. CP-IRL’s regret (blue) shrinks monotonically toward zero over the same range. Variance grows with u for both curves, consistent with tie-breaking sensitivity as $\hat{\theta}$ approaches the exact degenerate boundary.

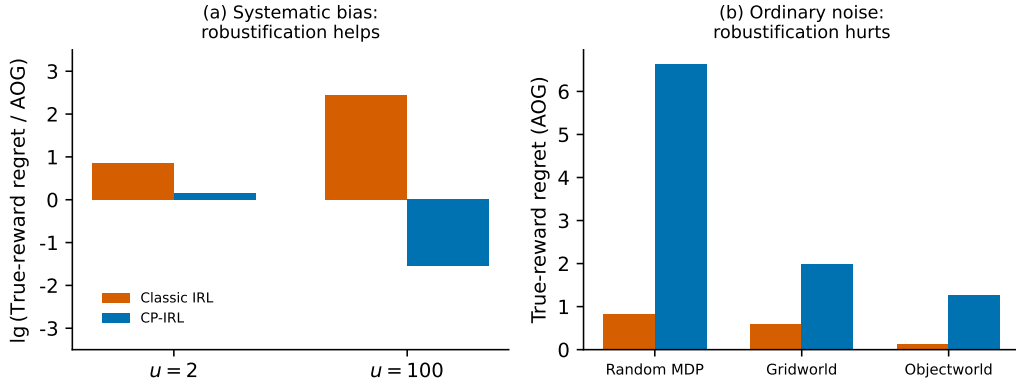


Figure 3: The Q2 scope condition. **(a)** On the degenerate construction of Theorem 4 (log scale), robust hedging cuts true-reward regret by 5–10,000 $\times$ as u grows from 2 to 100. **(b)** On three generic environments with ordinary, unbiased demonstrator noise (LP-IRL, $\gamma = 0.8$), robust hedging instead *costs* regret in every case. Same mechanism (worst-case hedging over a calibrated cap), opposite outcome, depending only on whether the point estimate is systematically biased.

333 angle responds sensibly to γ , demonstrator noise, and the train/calibration split. DKW is measurably
 334 looser than exact conformal calibration at matched confidence, everywhere tested. The calibrated cap
 335 is also far from vacuously wide: sweeping a *unified* noise parameter that controls every source of
 336 demonstrator-population variation at once, the calibrated angle shrinks monotonically from 27.7°
 337 to effectively 0° as that noise vanishes. Read through Proposition 2, this is a sweep of the induced
 338 regularization weight from $\tan 27.7^\circ \approx 0.53$ down to 0 – a moderate penalty for a genuinely
 339 heterogeneous population, and none at all for a homogeneous one, which is the qualitative behavior
 340 one would want of a hyperparameter chosen by hand.

341 **Q2: the benefit depends on systematic bias, not noise.** Figure 3 makes this concrete. On the
 342 degenerate construction, where the point estimate is biased by construction, robust hedging cuts
 343 true-reward regret sharply. On all three generic environments, under ordinary demonstrator noise,
 344 robust hedging *increases* regret relative to the plain estimate – consistently across all three point

estimators and across a full sweep of γ and the train/calibration split. This is not a solver bug: the robust solver’s worst-case value matches an independent second-order-cone-program oracle to solver precision, and its worst-case value is confirmably better than the plain estimate’s over the same calibrated set. It also is not a landscape-smoothness artifact: a 14° reward perturbation flips the optimal action at 10–13% of states across all three environments (9.6% Objectworld, 9.9% gridworld, 13.1% random MDPs), and Objectworld – measurably the least sensitive of the three – still shows robust hedging costing regret. Systematic bias in the point estimate, not policy-landscape smoothness, is what decides whether hedging helps.

Proposition 2 says why this had to be so: the hedge is a fixed tilt by $-\tan \alpha_\gamma b(\Phi^\top \mu)$ that never observes the realized error $\bar{\theta} - \theta_{\text{new}}^*$, so it pays for itself only when that error repeats. The degenerate construction supplies such an error – the LP estimator lands on θ_u by tie-breaking, identically every time – and ordinary demonstrator noise does not. These results are therefore a predicted boundary being confirmed, not an unexplained negative.

Q3: the conclusion holds across estimators, and separation is LP-specific. The gridworld comparison combines all three point estimators with all three decision rules: MaxEnt and Bayesian IRL recover an accurate point direction where the LP estimator has nontrivial error, yet in all three cases conformal robustification is less conservative than DKW but costs more true-reward regret than the corresponding point policy – the same pattern holds in Objectworld. We also ran MaxEnt and Bayesian IRL on the degenerate construction of Theorem 4 itself (N up to 3000, u up to 100, 10 seeds): neither estimator’s regret grows with u ; both stay at exactly zero.

The reason is structural, not that these estimators are simply more accurate. The LP estimator’s sub-optimality loss demands an exact fit to every observed decision, which forces it onto the degenerate tie direction θ_u whenever both decision classes appear in the population – an event of probability 1 as $N \rightarrow \infty$, for any finite u . MaxEnt and Bayesian IRL instead match an aggregate statistic (expected feature counts, or a Boltzmann likelihood) that converges by the law of large numbers to a fixed point set by the population’s class-split probabilities; landing exactly on the tied boundary would require that aggregate to hit a measure-zero direction, which does not happen along this construction. This narrows Theorem 4’s claim: the unbounded-regret failure mode belongs to *exact-fit* point estimation, of which LP sub-optimality loss is the leading IRL example, not to point estimation generally.

Together, Q1–Q3 separate two claims that are often conflated: conformal calibration can be tighter than a generic concentration bound, and optimizing for its worst case need not improve average performance under the true reward. The first is unconditionally true in our experiments; the second holds only under systematic bias.

7 Discussion and Limitations

When does CP-IRL help, precisely? Our theory (Theorem 4) proves a separation on a construction where the point estimate is *systematically* biased – classical point-estimate IRL’s regret diverges there while CP-IRL’s vanishes. Our experiments show that on generic environments with unbiased-but-noisy point estimates, it does not, and in fact costs regret in exchange for a worst-case guarantee that is real (independently verified) but not free. This is a precise, falsifiable scope condition, not a hedge: the method should be deployed when there is reason to believe point estimation is subject to a genuine structural bias rather than applied by default whenever demonstration data is merely noisy.

Proposition 2 explains the condition and takes a partial step toward acting on it. Because robustification here is a fixed, θ^* -independent tilt of the objective, it can only cancel error that is itself fixed; and because the tilt’s magnitude is computable before deployment, a practitioner can at least measure how large a bet they are placing, even without knowing whether it will pay. What remains open is the harder half: characterizing which real-world settings generate systematic rather than noisy estimation error, and diagnosing that distinction from data alone. Our $\tan \alpha_\gamma b(\Phi^\top \mu_{\text{plain}})$ bound measures the size of the wager, not its expected value.

What is not yet established. The general-case regret bound is loose – up to several hundred-fold on the one instance where we can check it against an exact result – so read it as a worst-case guarantee, not a typical-case predictor. A source-to-target probe shows a reward set calibrated on one MDP does not transfer safely to a related MDP with different dynamics: raising the gridworld’s slip probability from 0.10 to 0.15 at target coverage 0.8 drops mean coverage to 0.567 over eight seeds.

This establishes the failure of naive transfer, not a fix; combining covariate-shift weighting (Tibshirani et al. Tibshirani et al. [2019]) with the principal-angle transferability conditions of Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024] is future work.

We also tested the RLHF relabeling from §2: a “response selection” MDP where the population is annotators rather than task demonstrators, using the calibration and robust-MDP code unchanged. Coverage meets or exceeds target across $\gamma \in \{0.5, 0.7, 0.8, 0.9\}$ (0.947–0.994, 8 seeds, one configuration) – and, since one configuration and 8 seeds is thin evidence for a claim about a general property, we reran the same pipeline across four annotator-population configurations (4–10 responses, 3–6 attributes, 15–60 annotators) and 30 seeds at $\gamma \in \{0.7, 0.8, 0.9\}$: mean coverage across all twelve (configuration, γ) cells is 0.966–0.991, with at least 93% of seeds per cell meeting target, consistent with the marginal (not per-seed) nature of the conformal guarantee.

The gap above target likely comes from the same score-atom over-coverage as Theorem 2, sharpened by small annotator populations; the same systematic-bias-conditional benefit from Q2 applies here too. All experiments stay at tabular scale by design; we make no claim about function approximation or language-model settings.

8 Conclusion

IRL is inverse optimization with an MDP as its forward problem. That connection lets us transfer CIO’s calibrate-then-robustify principle to reward learning, given the sequential machinery it needs: a resolvent-based calibration score, a convex robust-MDP formulation for compact reward ambiguity, and coverage, regret, and separation guarantees.

Working out what that robust MDP computes turns out to be as informative as showing it is solvable. For the calibrated cap it is a regularized point-estimate MDP whose penalty weight, $\tan \alpha_\gamma$, the conformal quantile fixes on its own – an appealing property, since the robustness level comes with a finite-sample guarantee instead of a tuning sweep. But the same form shows the penalty is a fixed tilt that never sees the realized estimation error, so it can correct a systematic mistake and not a random one. Our experiments find exactly that: the deployed policy cuts true-reward regret under systematic bias and pays a premium under ordinary noise. Calibration certifies uncertainty; it does not make worst-case optimization free. That distinction, and the conditions under which each side of it holds, is what CP-IRL actually delivers.

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480 A Proofs

481 A.1 Proof of Proposition 1

482 The value function V_{θ}^{π} solves the linear Bellman equation:

$$V_{\theta}^{\pi} = \Phi_{\pi}\theta + \beta P_{\pi}V_{\theta}^{\pi}.$$

483 Since $\beta \in [0, 1)$ and P_{π} is a stochastic matrix, its spectral radius satisfies $\rho(\beta P_{\pi}) \leq \beta < 1$, so
484 $I - \beta P_{\pi}$ is invertible via a Neumann series, yielding:

$$V_{\theta}^{\pi} = (I - \beta P_{\pi})^{-1}\Phi_{\pi}\theta = W_{\pi}\theta.$$

485 Substituting V_θ^π into the Bellman equation for the action-value function $Q_\theta^\pi(s, a)$ gives:

$$\begin{aligned} Q_\theta^\pi(s, a) &= \phi(s, a)^\top \theta + \beta P(s, a, \cdot)^\top V_\theta^\pi \\ &= \phi(s, a)^\top \theta + \beta P(s, a, \cdot)^\top W_\pi \theta \\ &= \Psi_\pi(s, a)^\top \theta, \end{aligned}$$

486 where $\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot)^\top W_\pi$, which is linear in θ as claimed.

487 By the policy improvement theorem, π is Bellman-optimal for reward θ if and only if

$$Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \quad \forall s \in S, \forall a \in A.$$

488 Substituting the linear expression for $Q_\theta^\pi(s, a)$ yields exactly the stated feasible set:

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \quad \forall s, a\},$$

489 with no relaxation since the substitution is an equality.

490 The calibration score

$$c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$$

491 is then a linear objective over a set of $|S||A|$ linear inequalities plus the unit-ball constraint in $\theta \in \mathbb{R}^d$
492 alone – an SOCP of size d , independent of $|S|$ once Ψ_{π_k} is precomputed. ■

493 A.2 Proof of Theorem 1

494 **(a)** Define $g_C(x) := \min_{\theta \in C} \theta^\top x$. This is a pointwise infimum of the affine family $\{x \mapsto \theta^\top x : \theta \in C\}$. Since a pointwise infimum of affine functions is concave by definition for any nonempty C
495 (convex, nonconvex, or discrete – concavity of g_C requires no assumption on C 's shape), composing
496 g_C with the linear map $\mu \mapsto \Phi^\top \mu$ preserves concavity:

$$\mu \mapsto g_C(\Phi^\top \mu) = \min_{\theta \in C} \theta^\top \Phi^\top \mu$$

498 is concave in μ .

499 Since $M(P)$ is a fixed polytope that does not depend on θ (the transition kernel P is known and
500 fixed), the ambiguity set never touches the feasible region, only the objective. Therefore,

$$\max_{\mu \in M(P)} g_C(\Phi^\top \mu)$$

501 is a concave maximization over a fixed compact convex set, i.e., a convex program, for *any* nonempty
502 compact C . No Bellman recursion or rectangularity assumption enters this argument.

503 **(b)** Consider a 2-state, 2-action MDP with a scalar coupling parameter $t \in [0, 1]$ linking reward
504 $\theta(t) = 1 + t$ and transition probability $q(t) = 1 - t$. Let $p \in [0, 1]$ be the (scalar) probability
505 the agent's policy assigns to the action whose value is being computed. The induced worst-case
506 discounted value is:

$$W(p) = \min_{t \in [0, 1]} \frac{p(1+t)}{1 - \beta p q(t)} = \min_{t \in [0, 1]} \frac{p(1+t)}{1 - \beta p(1-t)}.$$

507 At discount factor $\beta = 0.9$, evaluating $W(p)$ at three points gives:

$$W(0) = 0,$$

$$W(5/9) = \min_{t \in [0, 1]} \frac{5(1+t)}{4 + 5t} = 1 \quad (\text{attained at } t = 0),$$

$$W(5/18) = \min_{t \in [0, 1]} \frac{10(1+t)}{31 + 5t} = \frac{10}{27} \approx 0.370 \quad \left(\text{attained at } t = 0, \text{ as } \frac{d}{dt} \frac{10(1+t)}{31+5t} = \frac{260}{(31+5t)^2} > 0 \right).$$

508 Since $5/18 = \frac{1}{2}(0 + 5/9)$, concavity of W would require:

$$W(5/18) \geq \frac{1}{2} [W(0) + W(5/9)] = \frac{1}{2}(0 + 1) = 0.5.$$

509 However, $0.370 < 0.5$, producing a strict Jensen violation with every quantity a ratio of small
510 integers, hand-verifiable without a solver.

511 Hence W is not concave, and the analogous worst-case value function for this jointly-ambiguous
512 (θ, P) instance is not concave either, so the outer maximization is not a convex program in this regime.

513 ■

514 A.3 Proof of Proposition 2

515 Fix $x \in \mathbb{R}^d$ and write $a = \bar{\theta}^\top x$, $x_\perp = x - a\bar{\theta}$, $b = \|x_\perp\|_2$, so $\|x\|_2^2 = a^2 + b^2$. Any θ with $\|\theta\|_2 \leq 1$
 516 decomposes as $\theta = u\bar{\theta} + v\hat{w}$ with $\hat{w} \perp \bar{\theta}$ a unit vector, $u = \bar{\theta}^\top \theta$, $v \geq 0$, and $u^2 + v^2 = \|\theta\|_2^2 \leq 1$.
 517 The cap constraint $\theta^\top \bar{\theta} \geq \cos \alpha$ reads $u \geq \cos \alpha$. Since $\theta^\top x = ua + v\hat{w}^\top x_\perp$ and $\hat{w}$ enters only
 518 through $\hat{w}^\top x_\perp \in [-b, b]$, the minimizing choice is $\hat{w} = -x_\perp/b$ when $b > 0$ (any $\hat{w}$ when $b = 0$),
 519 giving $\theta^\top x = au - bv$. The inner problem is therefore the two-variable program

$$\min \{ au - bv : u \geq \cos \alpha, v \geq 0, u^2 + v^2 \leq 1 \}.$$

520 As $b \geq 0$, it is optimal to take v maximal for the chosen u , i.e. $v = \sqrt{1 - u^2}$, leaving
 521 $\min_{u \in [\cos \alpha, 1]} f(u)$ with $f(u) := au - b\sqrt{1 - u^2}$ and $f'(u) = a + bu/\sqrt{1 - u^2}$, which is strictly in-
 522 creasing in u on $(-1, 1)$. Hence f is either increasing on the whole interval (the constraint $u \geq \cos \alpha$
 523 binds) or has a unique interior stationary point.

524 *Case 1:* $f'(\cos \alpha) \geq 0$, i.e. $a + b \cot \alpha \geq 0$. Then f is nondecreasing on $[\cos \alpha, 1]$ and the minimum
 525 is at $u = \cos \alpha$, $v = \sin \alpha$, giving $f = a \cos \alpha - b \sin \alpha$.

526 *Case 2:* $f'(\cos \alpha) < 0$. Setting $f'(u^*) = 0$ gives $u^*/\sqrt{1 - u^{*2}} = -a/b =: c > 0$, so $u^* =$
 527 $c/\sqrt{1 + c^2}$ and $f(u^*) = (ac - b)/\sqrt{1 + c^2} = -\sqrt{a^2 + b^2} = -\|x\|_2$, the unconstrained minimum
 528 over the unit ball, attained at $\theta = -x/\|x\|_2$. This is feasible exactly when $-x/\|x\|_2$ lies in
 529 the cap, i.e. $a \leq -\|x\|_2 \cos \alpha$, which is equivalent to $f'(\cos \alpha) < 0$ after squaring. The two
 530 cases agree at the boundary $a = -\|x\|_2 \cos \alpha$, where $b = \|x\|_2 \sin \alpha$ and $a \cos \alpha - b \sin \alpha =$
 531 $-\|x\|_2(\cos^2 \alpha + \sin^2 \alpha) = -\|x\|_2$, so the stated piecewise formula is continuous.

532 For the second claim, $a(\Phi^\top \mu) \geq 0 > -\|\Phi^\top \mu\|_2 \cos \alpha$ places every $\mu \in M(P)$ in Case 1, so the
 533 robust objective is $\mu \mapsto a \cos \alpha - b \sin \alpha$ uniformly on $M(P)$. Since $\alpha < \pi/2$ gives $\cos \alpha > 0$,
 534 dividing the objective by the positive constant $\cos \alpha$ leaves the $\arg \max$ unchanged and yields
 535 $a - \tan \alpha b$, which is the stated program. Concavity is immediate from this form: $a(\Phi^\top \mu) = \bar{\theta}^\top \Phi^\top \mu$
 536 is linear in μ , while $b(\Phi^\top \mu)$ is a seminorm composed with a linear map, hence convex, and $\tan \alpha \geq 0$.
 537 ■

538 A.4 Proof of Theorem 2

539 Under Assumption 1, the sequence of pairs

$$(\hat{\theta}_1, \pi_1), \dots, (\hat{\theta}_{N_{\text{val}}}, \pi_{N_{\text{val}}}), (\hat{\theta}_{\text{new}}, \pi_{\text{new}})$$

540 are i.i.d., and therefore exchangeable. The calibration score

$$c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$$

541 (Proposition 1) is a fixed measurable function of $(\hat{\theta}_k, \pi_k)$ once $\bar{\theta}$ is fixed (computed on a disjoint
 542 training split, independent of the calibration and test data). Consequently, the calibration scores

$$c_1, \dots, c_{N_{\text{val}}}, c_{\text{new}}$$

543 are themselves exchangeable.

544 By the standard rank argument for exchangeable sequences, the rank of c_{new} among all $N_{\text{val}} + 1$
 545 scores is uniform on $\{1, \dots, N_{\text{val}} + 1\}$ (up to ties), which implies:

$$\mathbb{P}(c_{\text{new}} \geq c_{(\tau)}) \geq \frac{\tau}{N_{\text{val}} + 1} \geq \gamma$$

546 for $\tau = \lceil (N_{\text{val}} + 1)\gamma \rceil$.

547 It remains to show that the event $\{c_{\text{new}} \geq c_{(\tau)}\} = \{c_{\text{new}} \geq \cos \alpha_\gamma\}$ is equivalent to $\hat{\Theta}_{\text{new}} \cap$
 548 $C(\bar{\theta}, \alpha_\gamma) \neq \emptyset$:

$$\begin{aligned} c_{\text{new}} \geq \cos \alpha_\gamma &\iff \exists \theta \in \hat{\Theta}_{\text{new}} \text{ such that } \theta^\top \bar{\theta} \geq \cos \alpha_\gamma \\ &\iff \exists \theta \in \hat{\Theta}_{\text{new}} \text{ such that } \theta \in C(\bar{\theta}, \alpha_\gamma) \\ &\iff \hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset. \end{aligned}$$

549 Combining these statements yields:

$$\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma. \quad \blacksquare$$

550 A.5 Proof of Theorem 3

551 Write

$$V_\theta(\mu) := \frac{\theta^\top \Phi^\top \mu}{1 - \beta}$$

552 for the expected discounted return of occupancy measure μ under reward θ . By Cauchy–Schwarz,
553 this function is $\bar{\nu}$ -Lipschitz in θ (in ℓ_2) uniformly over $\mu \in M(P)$:

$$\begin{aligned} |V_\theta(\mu) - V_{\theta'}(\mu)| &= \frac{|(\theta - \theta')^\top \Phi^\top \mu|}{1 - \beta} \\ &\leq \|\theta - \theta'\|_2 \frac{\|\Phi^\top \mu\|_2}{1 - \beta} \\ &\leq \|\theta - \theta'\|_2 \nu(\mu). \end{aligned}$$

554 By the almost-sure-covering hypothesis, there exists a reward vector

$$\theta_1 \in \hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_1) \quad \text{almost surely.}$$

555 By the population-mean hypothesis, $\|\bar{\theta} - \theta_{\text{new}}^*\|_2 \leq \sigma$. By the bounded-diameter hypothesis,
556 $\|\theta_1 - \theta_{\text{new}}^*\|_2 \leq \eta$ (since both θ_1 and θ_{new}^* are optimal for policies within the same feasible-set
557 diameter bound around the true optimum).

558 By the chord-length identity for unit vectors at angle ψ , $\|\theta - \theta'\|_2 = 2 \sin(\psi/2)$. Since any two
559 points within $C(\bar{\theta}, \alpha_1)$ are within angle $2\alpha_1$ of each other:

$$\|\theta_1 - \bar{\theta}\|_2 \leq 2 \sin \alpha_1.$$

560 By the triangle inequality:

$$\|\theta_1 - \theta_{\text{new}}^*\|_2 \leq \|\theta_1 - \bar{\theta}\|_2 + \|\bar{\theta} - \theta_{\text{new}}^*\|_2 \leq 2 \sin \alpha_1 + \sigma.$$

561 The robust policy’s minimax optimality guarantees $V_{\theta_1}(\mu_{\text{CIRL}}) \geq V_{\theta_1}(\mu^*)$ (since $\theta_1 \in C(\bar{\theta}, \alpha_1)$
562 and the robust policy maximizes worst-case return over this set). We then decompose the average
563 optimality gap:

$$\begin{aligned} \text{AOG} &= V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}}) \\ &\leq [V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_1}(\mu^*)] + [V_{\theta_1}(\mu_{\text{CIRL}}) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}})], \end{aligned}$$

564 where we used $V_{\theta_1}(\mu^*) \leq V_{\theta_1}(\mu_{\text{CIRL}})$ to drop the middle terms.

565 Bounding each bracket using Lipschitz continuity:

$$\begin{aligned} V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_1}(\mu^*) &\leq \|\theta_1 - \theta_{\text{new}}^*\|_2 \nu(\mu^*) \leq (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu^*), \\ V_{\theta_1}(\mu_{\text{CIRL}}) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}}) &\leq \|\theta_1 - \theta_{\text{new}}^*\|_2 \nu(\mu_{\text{CIRL}}) \leq (\eta + \sigma) \nu(\mu_{\text{CIRL}}). \end{aligned}$$

566 Summing these two bounds gives:

$$\text{AOG} \leq (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu^*) + (\eta + \sigma) \nu(\mu_{\text{CIRL}}).$$

567 Substituting the uniform bound $\bar{\nu}$ for both $\nu(\mu^*)$ and $\nu(\mu_{\text{CIRL}})$ yields the fully distribution-free
568 corollary:

$$\text{AOG} \leq (2 \sin \alpha_1 + 2\eta + 2\sigma) \bar{\nu}. \quad \blacksquare$$

569 A.6 Proof of Theorem 4

570 The degenerate MDP’s occupancy-measure polytope $M(P)$ is the segment between the two actions’
571 feature vectors, parameterized by $p \in [0, 1]$ (the probability mass on the “risky” action). The robust
572 MDP reduces to the 2-point minimax game:

$$\max_{p \in [0, 1]} \min_{\theta \in C(\theta_u, \alpha)} \psi(\theta; p),$$

573 where

$$\psi(\theta; p) := p \theta^\top \phi_0 + (1 - p) \theta^\top \phi_1$$

574 is bilinear in (p, θ) , and

$$C(\theta_u, \alpha) = \{\delta \in [\delta_u - \alpha, \delta_u + \alpha]\} \quad \text{with } \theta = (\cos \delta, \sin \delta)$$

575 is a compact convex arc centered at δ_u .

576 Since both feasible sets are compact and convex and ψ is bilinear, Sion's minimax theorem applies:

$$\max_{p \in [0,1]} \min_{\theta \in C(\theta_u, \alpha)} \psi(\theta; p) = \min_{\theta \in C(\theta_u, \alpha)} \max_{p \in [0,1]} \psi(\theta; p).$$

577 At a saddle point $(\theta^\dagger, p^*)$, necessity of the marginal optimality conditions forces $\theta^\dagger$ to make p
 578 non-unique in the inner $\arg \max_p$ of an affine-in- p objective. This occurs only at a tie:

$$\theta^{\dagger \top} \phi_0 = \theta^{\dagger \top} \phi_1 \implies \delta^\dagger = \delta_u,$$

579 which is always interior to the arc since the cap is centered at δ_u .

580 Stationarity of $\delta \mapsto \psi(\delta; p^*)$ at δ_u requires:

$$-p^* \cos \delta_u + (1 - p^*) \sin \delta_u = 0.$$

581 Substituting $\cos \delta_u = \frac{1}{\sqrt{1+u^2}}$ and $\sin \delta_u = \frac{u}{\sqrt{1+u^2}}$ and simplifying yields:

$$p^*(u) = \frac{u^2}{1 + u^2},$$

582 which is independent of α .

583 A second-order check confirms this is a genuine minimum:

$$\left. \frac{\partial^2}{\partial \delta^2} \psi(\delta; p^*) \right|_{\delta=\delta_u} = \frac{u}{\sqrt{1+u^2}} > 0 \quad \forall u > 0.$$

584 Thus, the robust optimum is a genuine mixture placing weight $p^*(u) \rightarrow 1$ (as $u \rightarrow \infty$) on the risky
 585 action. The resulting robust value stays bounded and the true regret shrinks to zero because the
 586 calibrated cap's worst case is evaluated near θ_u (not the true θ^*), and the mixture hedges against that
 587 local worst case.

588 By contrast, the classical point estimate converges to the pure degenerate tie-break (θ_u) with no such
 589 hedge, yielding regret that grows linearly in u under the realized tie-breaking convention ($p_{\text{classic}} = 0$
 590 deterministically, the pure risky vertex). ■